utomatic Paired Trajectory Construction

For ML researchers and NLP engineers in tool-augmented LLM and RLHF systems

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs must choose among three actions each turn:

- Ask follow-up questions for missing information
- Execute tool call when slots are complete
- Reject tool invocation for out-of-scope requests

SFT-only failure modes:

- Missing follow-up questions → premature/hallucinated tool calls
- Incomplete slot handling → low completion quality
- Weak rejection behavior → irrelevant tool usage

Motivation

SFT expert trajectories capture what to say, but not robust turn-level control over when to ask, call, or reject tools.

DiaTool-DPO adds preference learning for dialogue flow decisions.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

State semantics:

- S1 Initial: parse user request and determine tool relevance
- S2 Partial slots: ask clarifying questions (can repeat)
- S3 Complete slots: ready for tool execution
- S4 Await tool response before final reply
- S5 Complete: finish conversation or task

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Brief Example
Type 1	1→3→4→5	User provides all required slots upfront; model calls tool directly.
Type 2	1→(2*N)→3→4→5	User omits key slot(s); model asks follow-ups until complete.
Type 3	1→1	Request is out-of-scope; model rejects tool call and responds safely.

These trajectories define how chosen/rejected dialogue pairs are generated for DiaTool-DPO training.

Automatic Paired Trajectory Construction Without Human Labels

For each query type, construct chosen vs. rejected trajectories automatically:

- Type 1: chosen uses direct tool call; rejected introduces unnecessary slot-filling
- Type 2: chosen asks clarifying questions; rejected hallucinates missing slots
- Type 3: chosen rejects tool call; rejected performs inappropriate invocation

Result: preference pairs encode dialogue control policies
without costly human pairwise annotations.

Type	#Pairs	Difficulty
Type 1	64K	Easy
Type 2	64K	Hard
Type 3	64K	Easy

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.243	0.570	0.389	0.522	0.426	0.431
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	0.886	0.929	0.833	0.826	0.884	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	0.743	0.871	0.833	0.826	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	0.857	0.929	0.917	0.913	0.905	**0.904**

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Key gains on LLaMA-3-8B:

- Slot: 0.639 → 0.917 (+43.5%)
- Relevance: 0.826 → 0.913 (+10.5%)

Interpretation:
DPO significantly improves slot acquisition and out-of-scope/tool-rejection quality.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1) 5-state MDP gives a principled control view of tool-augmented dialogue.
2) Three query trajectories enable automatic chosen/rejected pair generation.
3) SFT + DiaTool-DPO yields strongest balanced performance (best Macro Avg across models).
4) DPO complements, not replaces, SFT (DPO-only underperforms).
5) Largest gains appear in slot-filling and tool rejection; method is language-agnostic.
6) Practical path toward near GPT-4o-level tool dialogue quality on smaller open models.